CDT 18V-100/400A 4CH
Charge/Discharge Test System

USER MANUAL
REV. A

Pi Em Si Ltd

Dobri Chintulov Str. 9,
Sliven 8800, Bulgaria
Tel: (359) 044 62 47 91
Fax: (359) 044 62 48 59
office@pmc-bg.com, www.pmc-bg.com

Contents

Chapter 1	**DECLARATION OF CONFORMITY**	**4**
Chapter 2	**WARRANTY**	**5**
Chapter 3	**SAFETY INSTRUCTIONS**	**6**
Chapter 4	**GENERAL INFORMATION**	**10**
	USER MANUAL CONTENT	**10**
	INTRODUCTION	**10**
Chapter 5	**TECHNICAL DATA AND SPECIFICATIONS**	**12**
Chapter 6	**INSTALLATION**	**15**
	GENERAL	**15**
	PREPARATION OF USE	**15**
Chapter 7	**OPERATING INSTRUCTION**	**21**
	MANAGEMENT AND DISPLAY CONTROLS	**21**
	MAIN SCREEN	**22**
	LOG SETTINGS SCREEN	**24**
	ALARM MESAGES	**25**
	OPERATION	**26**
Chapter 8	**MAINTENANCE**	**27**
	INTRODUCTION	**27**
	UNITS UNDER WARRANTY	**27**
	PERIODIC MAINTENANCE	**27**
	ADJUSTMENTS AND CALIBRATION	**27**
	PARTS REPLACEMENTS AND REPAIRS	**27**
	SOFTWARE UPGRADE	**28**
	TROUBLESHOOTING	**28**

DECLARATION OF CONFORMITY

1

PI EM SI Ltd
Dobri Chintulov Str.9
Sliven 8800, Bulgaria
Tel.: +359 888 875 269
e-mail:office@pmc-bg.com

DECLARATION OF CONFORMITY
CDT 18V / 50A

We, PI EMSILtd,of Dobri Chintulov Str.9, Sliven8800, Bulgaria declare under our sole responsibility that the charge / discharge testers: CDT18V/50A comply with the provisions of the folowing European Directive and are eligible to bear the CE mark:

Low Voltage Directive: 2006/95/EC.

Assurance of conformance of the described products with theprovisions of the stated EC directive isgiven through compilance to the following standard:

Electricical Safety IEC EN 61010-1 3rd Edition

Name of Authorized Signatory	Valentin Petrov Vandev
Signature of Authorized Signatory	
Position of Authorized Signatory	Managing director, PI EM SI Ltd, Sliven, Bulgaria
Date	9 February 2018
Place where signed	Sliven Bulgaria

WARRANTY 2

This product is warranted against defects in materials and workmanship for a period of thirty months from date of shipment. During the warranty period, Pi Em Si Ltd. will, at it's option, either repair or replace products which prove to be defective.

LIMITATION OF WARRANTY

The warranty shall not apply to defects resulting from locating and installing the testers that do not conform to requirements written in this manual.

The warranty shall not apply to defects resulting from improper or inadequate usage or maintenance by the buyer.

The warranty shall not apply to defects resulting from unauthorized modifications, or from operation exceeding the environmental specifications of this product.

WARRANTY SERVICE

This product must be returned to a Pi Em Si Ltd. service facility for repairs or other warranty service. For products returned to Pi Em Si Ltd. for warranty service, the buyer shall prepay shipping charges to Pi Em Si Ltd. If the unit is covered under the foregoing warranty then Pi Em Si Ltd. shall pay the shipping charges to return the product to the buyer.

SAFETY INSTRUCTIONS 3

CAUTION

The following safety precaution must be observed during all phases of operation, service and repair of this equipment. Failure to comply with the safety precautions or warnings in this document violates safety standards of design, manufacture and intended use of this equipment and may impair the built in protections within. Pi Em Si Ltd. shall not be liable for user's failure to comply with these requirements.

GROUNDING

The testers are mounted inside a metal cabinet(delivered by Pi Em Si Ltd.) which must be connected to the AC power supply mains through a five conductor power cable (32 A, 5 pin plug - 3P,N,E). To minimize shock hazard, the cabinet's chassis must be connected to an electrical ground. In both cases, any interruption of the protective ground conductor willcause a potential shock hazard that might cause personal injury.



Grounding of the output cables of the testers is unacceptable.

FUSES

Fuses mounted inside the tester must be changed by authorized Pi Em Si Ltd. service personnel only. For continued protection against risk of fire, replace only with the same type and rating of the damaged fuse.
Fuses mounted on the voltage feedback cables can be replaced by byuer's personnel, following the recommendation for using the same type and rating.

INPUT RATINGS

When used standalone, tester's power supply is within 220 V / 50-60 Hz , with tolerance +10/-15 %.

If mounted inside a metal cabinet (delivered by Pi Em Si Ltd.) then the cabinet's power supply is within 3 x 380 V / 50-60 Hz, +10/-15 %.

Do not use AC supply, which exceeds the above input voltage and frequency ranges.

PARTS SUBSTITUTIONS & MODIFICATIONS

Parts substitutions and modifications are allowed by authorized Pi Em Si Ltd. service personnel only. Repairs and modifications of the testers can be performed only by personnel of Pi Em Si Ltd.

Do not replace components with power supply turned on. To avoid injuries, always disconnect power and batteries, before taking any action of replacing components.

CONNECTING TO BATTERIES

Connecting to batteries must be performed only when the tester is turned off and the output cables are connected to the tester first.

Never disconnect batteries when a test program is running!

Do not connect batteries with a voltage exceeding the following output voltage range :

- for CDT 18V/50 A - 18 volts.

OUTPUT CABLES

Section of the power output cables shall not be less than 25 $mm^{2.}$ Do not use cables with corroded cable shoes and damaged insulation.

MEASUREMENT CABLES

The measurement cables are fuse protected (630 mA). If the fuse is damaged this means that measurement cables must be replaced.

Usage of fuse protected measurement cables is mandatory.

LOCATION, INSTALLATION AND VENTILATION

In discharge mode the energy is released as heat. There is fan cooling and the stream of cooling air is directed from the front to rear.

For the cooling to be effective, there must be at least 0.8 m. distance between the front and the rear of the testers, and any objects or walls.

Connecting and arranging cables should be done in such a way that they do not impede the flow of cooling air.

When in one room there are many testers it is necessary to provide air conditioning or ventilation.

ENVIRONMENTAL CONDITIONS

The CDT 18V/50A charge / discharge tester safety approval applies to the following operating conditions:

- indoor use;
- altitude up to 2000 m;
- temperature 5 °C to 40 °C;
- maximum relative humidity 80 % for temperatures up to 31°C decreasing linearly to 50 % relative humidity at 40 °C;
- mains supply voltage fluctuations up to +10/-15 % of the nominal voltage;
- pollution degree 2;

SAFETY INSTRUCTIONS 3

	Indicates ground terminal
I	On Supply
O	Off Supply
	CAUTION Risk of Electrical Shock
WARNING	The WARNING sign denotes a hazard. An attention to a procedure is called. Not following procedure correctly could result in personal injury. A WARNING sign should not be skipped and all indicated conditions must be fully understood and met.
CAUTION	The CAUTION sign denotes a hazard. An attention to a procedure is called. Not following procedure correctly could result in damage to the equipment. Do not proceed beyond a CAUTION sign until all indicated conditions are fully understood and met.

WARNING : Do not install the testers in an explosive environment. It is not recommended testers to be in the same room with the the batteries.

CAUTION : Incorrect test program can be a safety risk. Caution must be taken and only qualified personnel should create them.

GENERAL INFORMATION 4

USER MANUAL CONTENT

This User's Manual contains :

- Technical data for the CDT 18V/50A charge/discharge tester;
- Installation instruction;
- Operating instruction;
- Maintenance instruction;

This manual does not include a description of the option to manage and monitor the testers by PC. Information about this capability can be obtained in the operating manual for the relevant computer applications : "Battery Tester Manager v1.0" for the CDT 18V/50A tester.

INTRODUCTION

CDT 18V/50A are testers with microprocessor control and are designed for testing 1 to maximum of 6 cells with total voltage up to 18 V . This type of testers have the option for connecting to external data logging system for measuring the parameters of each cell of the battery (voltage, reference voltage, temperature, pressure).

For this tester there is an auxiliary application software for PC that allows: creating, editing, storing and loading of test programs; control and visualization of the testing process; maintain records of all tests performed and for their subsequent review in tabular and graphical form, and export to Excel file format.

The CDT 18V/50A can be used standalone without being connected to a PC. Test programs can be exported from the "Battery Test Manager" application to a flash drive (FAT or FAT32 formatted), and subsequently loaded and executed from the tester. Process data can be logged on the usb flash drive in the CSV file format and then loaded from EXCEL or similar data analysis application.

Main features

- charge and discharge;
- supports the following operating modes :
  - constant current - CC;
  - constant voltage - CV;
  - constant power - CP;
- capability of working in parallel
  - up to 12 circuits (6 if data logger is used) ;

- built-in TFT touch screen display visualizing : current program and step, output current and voltage, amper and watt hours, power, program work time, program status etc;
- buffer memory for storing process data and events : up to 520 sample records;
- programmable sample time for every step;
- local and remote control (Start , Stop, Resume, End, etc.);
- real time clock with automatic adjustment from the PC;
- built-in protections : overcurrent; overheat; current , voltage and power deviation;
- integrated galvanically isolated RS 485 serial interface that allows connection to PC;
- integrated galvanically isolated RS 485 serial interface for parallel mode and connection to external data logging system;
- integrated USB 2.0 interface for data logging and test program uploading;
- integrated interface for PC connection (sending process data to terminal software, and for programming the touch display);
- calibration using a system of parameters set by using the touch screen display;
- Automatic resume of a test program in case of interrupted power supply and its subsequent restoration;

Accessories

The following accessories are shipped with the products :

- converter RS 485/RS232 with cable for connection with PC;
- twisted pair cable for connecting the converter with testers;
- termination resistor 120 ohms, mounted on connector RJ 45;
- on request : external data logging system, cable sets, metal cabinet.

TECHNICAL DATA AND SPECIFICATIONS 5

CDT 18V/100A have the features, described in the following table :

Feature types	CDT 18V/100A
Input characteristics	
Power supply voltage	220 V / 50-60 Hz +10%/-15%
Power	< 1100 VA
Output characteristics	
Circuits	1
Output current in charge and discharge mode	0-50 A
Max.output power in charge mode	900 W
Max.output power in discharge mode *Depends on ambient temperature*	650 W (t <=25°C) 550 W (t<=35°C) 450 W(t<=40°C)
Output voltage in charge mode	0 - 18 V
Output voltage in discharge mode	1.5 - 16 V (min.voltage depends on power cables length, 1.5V - 5m.)
Cells in output circuit	depends on battery type 1-6 for lead acid batteries
Output cables	>= 25 mm^2 length < 10 m.

TECHNICAL DATA AND SPECIFICATIONS 5

Feature types	CDT 18V/100A
Analog inputs	
Battery voltage measurement	1 ch / range : -5V to +20 V / res.: 1 mV
Potentials measurement with reference electrode	1 ch / range : -5V to +5V / res. : 1 mV input resistance : 4 Mom
Temperature	1ch / range : -50 to +200°C Sensor type : Pt 100, 4 wire
Support for external data logging system	Yes
Work modes	
Charge	(CC, CV,CP)*
Discharge	(CC, CP)*
Parallel mode support	yes (up to 12 circuits)
Protections Over current, Over heat, Output current deviation, Output voltage deviation, Output power deviation, Power cables reversed polarity	
Resolution	
Current	0.001 A
Voltage	0.001 V
Accuracy	
Current	(20-100%) +/- 0.5 % Set Value (1-20%) +/- 0.1% Full Scale
Voltage	+/- 0.1 % Full Scale

Feature types	CDT 18V/100A
PC software "Battery Tester Manager"	
Communication	
Interfaces	one RS 485 - PC Control one RS 485 - Parallel mode and data logging two USB 2.0 ports for mass storage device (flash drive) and PC connection (terminal software)
Max. number of testers	up to 32 devices
Design parameters	
Dimensions (width mm. x depth mm. x height mm.)	Standalone tester : 429 x 508.5 x 138 Cabinet for 12 testers : 602.6 x 702.5 x 1948
Weight (kg.)	tester : 24 kg cabinet : 162 kg
Cooling - forced air cooling, directed from front to rear	

* CC -Constant Current CV - Constant Voltage CP - Constant Power

INSTALLATION 6

GENERAL

This chapter contains instructions for initial inspection, preparation for use and repackaging for shipment of the equipment.

Actions for providing PC connectivity and setting up a communication port are described in a separate chapter.

PREPARATION OF USE

To be operational the CDT 18V/50A tester must be connected to a suitable power supply. The power supply must comply with the specifications of the equipment.

Do not turn on power before reading the following instructions!

Initial check

Before being sent, the equipment has been checked and no mechanical or electrical defects were found. When unpacking the equipment upon receipt, it is necessary to look for any damage that may occur during shipment. The check should confirm that there is no external damage like broken buttons and connectors, and that the TFT touch display is not scratched or cracked. Also, ensure that all necessary screws are tight and secure. In case a damage is found, immediately notify the manufacturer.

Location, mounting and ventilation

It is mandatory to comply with the following distances from the testers to the surrounding objects and walls:

- front : no less than 0.8 m;
- rear : no less than 0.4 - 0.6 m.

The flow of cooling air is directed from the front to rear.

It is necessary to maintain the temperature of the environment (in the room) within the limits specified in the conditions of the environment in the safety instructions.

If a cabinet is used, the units should be properly fixed by using the provided plates on the left and right side of the cabinet. Two screws are used for each plate.

INSTALLATION 6

Power supply requirements

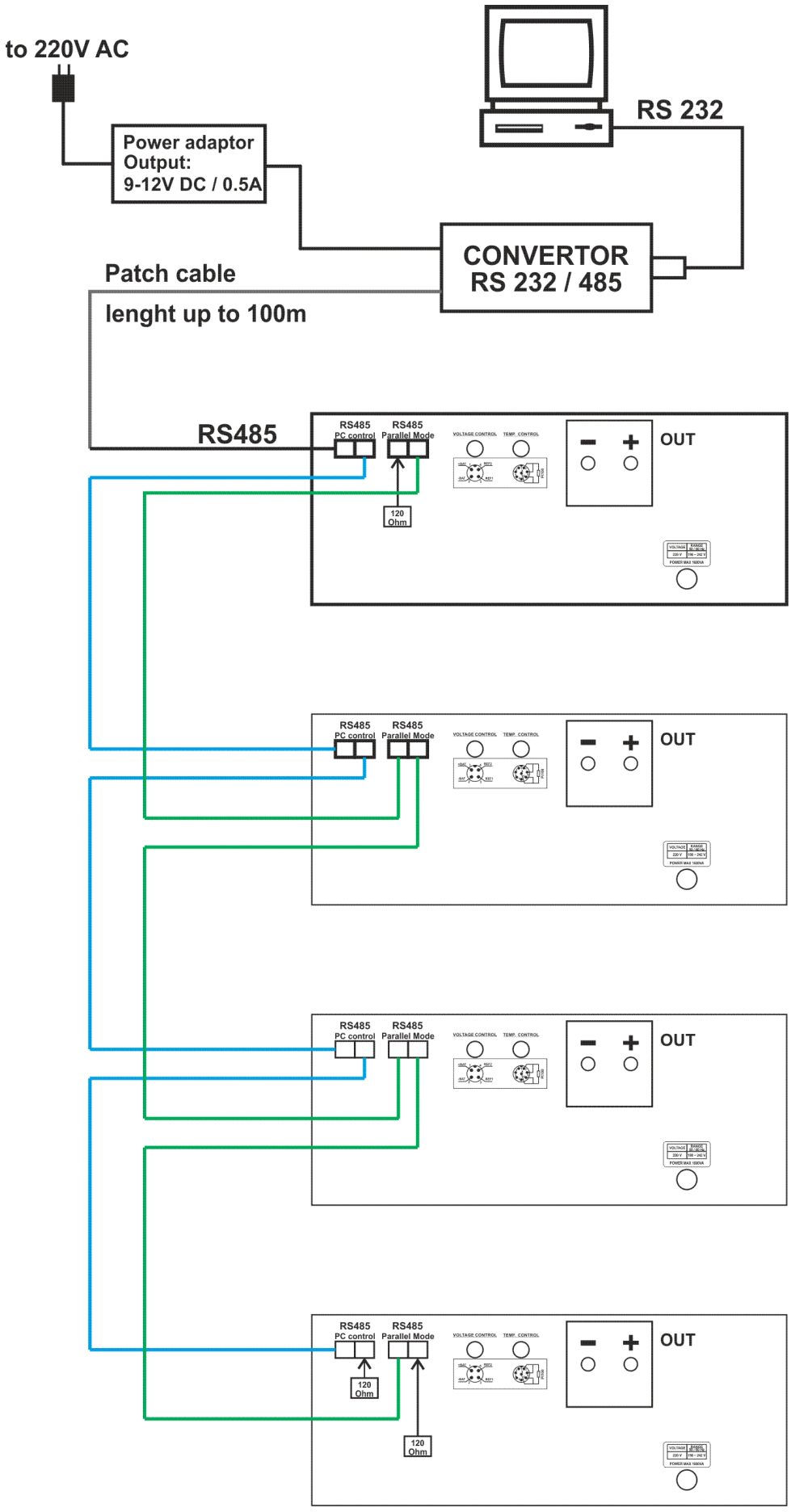
Standalone tester run on single phase power 220 V / 50-60 Hz. If mounted inside a metal cabinet (delivered by Pi Em Si Ltd.) then the cabinet's power supply is within 3 x 380 V / 50-60 Hz, +10/-15 %. Make sure that the mains power supply can provide power specified in the specifications within acceptable tolerance.

Connecting to power supply

Use for connecting to the power supply conductors with a cross section not less than that specified in the specification. When powering up a cabinet (with up to 12 testers) the cables must be arranged in a way that does not interfere with the cooling air flow.

Connecting to PC

It is performed by a RS485 / RS232 converter connected to a PC serial port, using a standard cable. The connection between the testers and the converter is by using standard patch cables, CAT .5E. The connectors for connecting to the PC are placed on the rear side of the testers and are marked (RS 485 PC Control) .


to 220V AC
Power adaptor
Output:
9-12V DC / 0.5A
RS 232
CONVERTOR
RS 232 / 485
Patch cable
lenght up to 100m
RS485
RS485
PC control
RS485
Parallel Mode
120
Ohm
OUT

To one communication channel can be connected up to 32 testers.
At the end of the line a termination resistor is placed.

Inside the RS485/RS232 converter there are termination resistors built in. The converter requires external power supply (9-12V DC 0.5A). A power adapter is shipped with the converter.

If the number of devices exceeds the aforementioned limits then a converter supporting the necessary number of communication channels must be used.

Parallel mode and external data logger connection

The connection between testers working in parallel (and data logger for CDT 18V/50A, if available) is performed using separate serial interface RS485. Standard patch cables, CAT .5E are used. Termination resistors are placed at both ends of the communication line. If data logger is used, it has a built in termination resistor.

Notes :

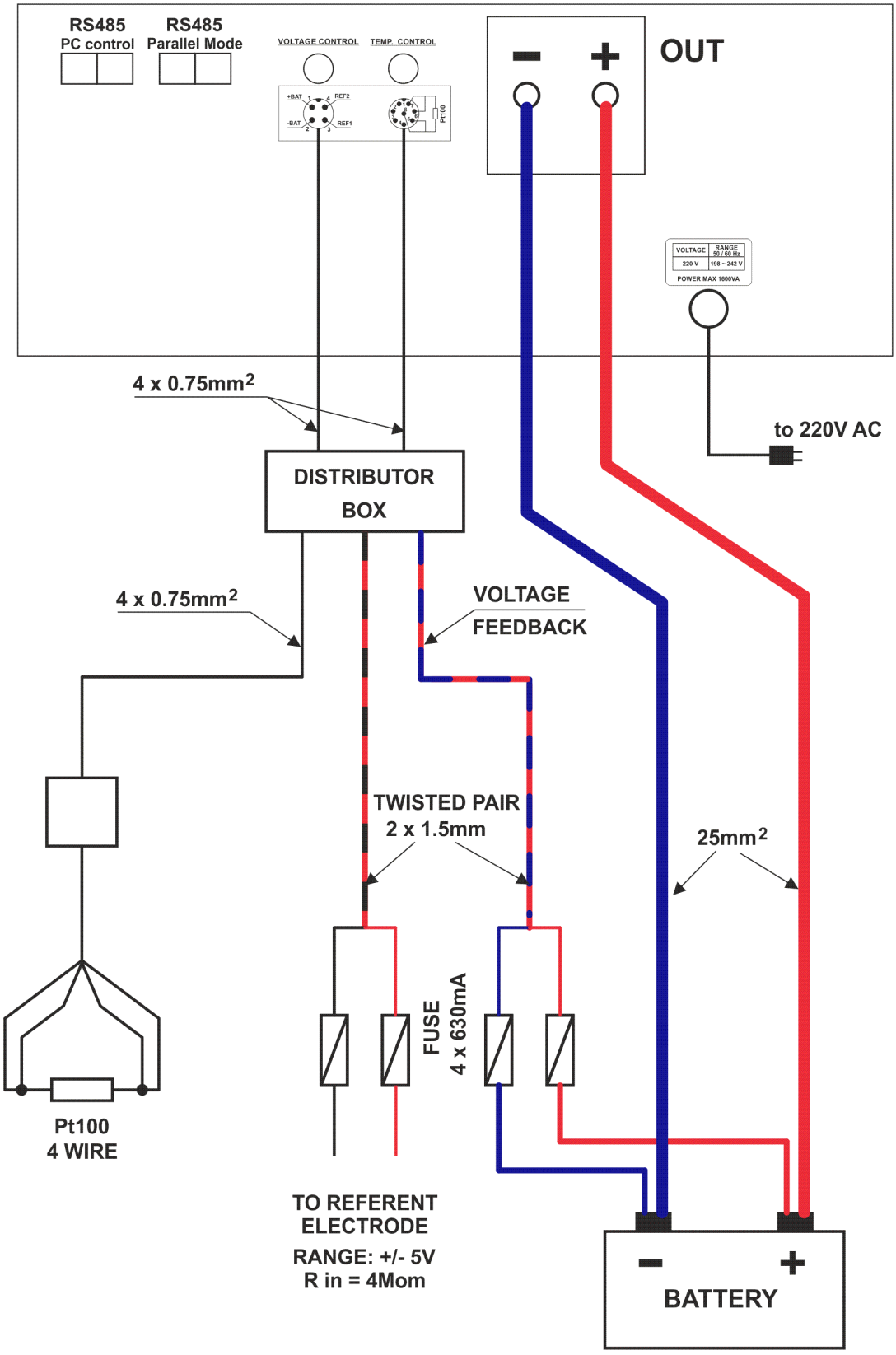
Only devices placed in the same rack may be grouped to work in parallel. Only one group of devices from a single rack is allowed.
Connecting multiple devices to the same communication line is a necessary but not sufficient condition for work in parallel. The needed devices must also be selected and grouped in the PC application software. Also, the power output cables must be connected to the battery.

First time connection of output and measurement cables

The following subsequent operations are executed before power is applied to the testers.

1. Plug the connectors of the measurement cables to the testers;
2. Mount the output power cables to the testers and put on the protection covers;
3. The output and measurement cables have to be marked. Make sure the marks are correct;
4. Connect the power supply cables of the testers to the sockets located at the right side of the rack (looked from the rear side);
5. Arrange cables in such a way that they do not impede the flow of cooling air;
6. Connect the rack power supply to the mains.

Connecting to batteries



INSTALLATION 6

Connecting the output and measurement cables to the batteries must be performed according to their marks. Also, care must be taken to not allow loose connections, as they can be reason for heating and fire.

IMPORTANT : the voltage feedback cables must be connected! Without them the tester will start in Constant Current mode, which for Li Ion batteries is dangerous!

Note : When working in parallel mode the power output cables are connected in parallel to the battery. Also, the measurement cables of all the circuits must be connected.

First test

Before executing the following steps, please read the next chapter first.

1. First, verify that you have followed the above instructions and everything is located, mounted and connected right;
2. Switch on the testers;
3. Start the "Battery Test Manager" PC application software ;
4. If there is no connection to the testers, select the respective serial port.
5. After connection between the PC and the testers is established a test program should be created, loaded and started for each tester in order to check if they work correctly.

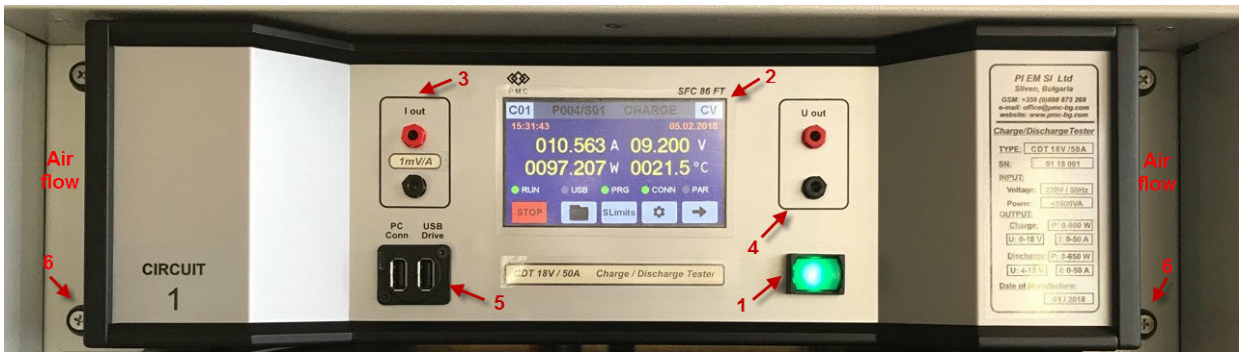
Note : 3) , 4) and 5) are explained in more detail in the operating manual for "Battery test manager v1.0" .

OPERATING INSTRUCTION 7

The CDT 18V/50 charge-discharge testers have a full set of controls, indicators and connectors that allow the user to easily setup and operate the unit. Before starting to operate the unit, please read the following sections for explanation of the functions of the controls and connectors terminals.

MANAGEMENT AND DISPLAY CONTROLS

Following is an image of the testers front side.



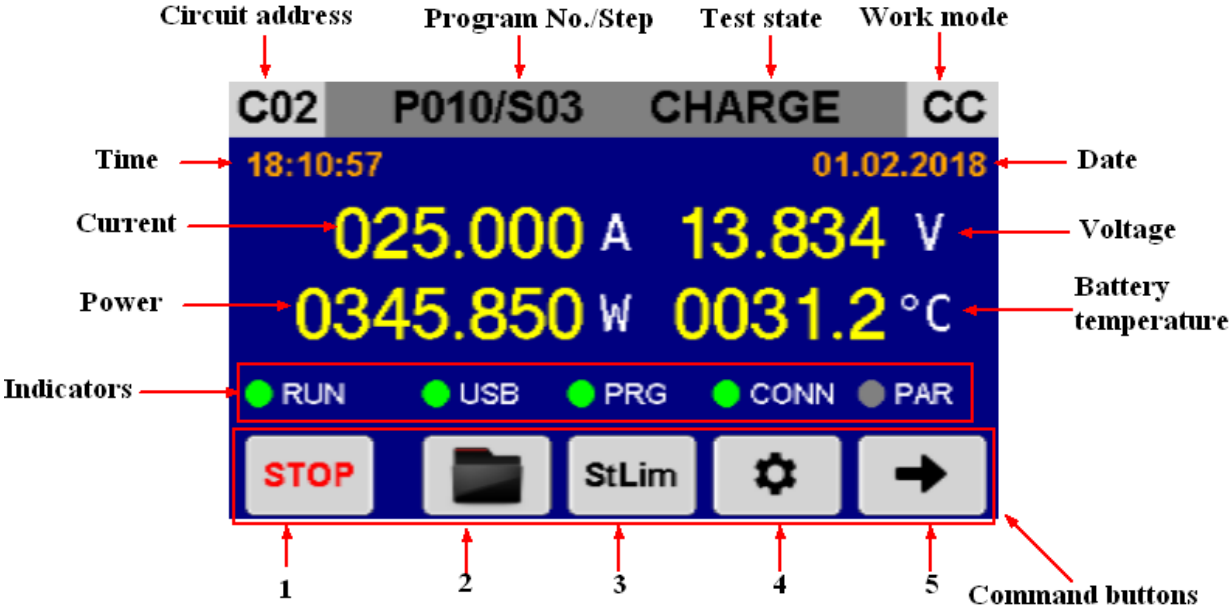
1. Main power push button with light indication ;
2. TFT touch display - has the following functions :
- indicates process data and test status;
- allows for different test management commands : "START", "STOP", "RESUME", "END";
- can be used for adjustments and calibration;
- simpfied manual program (for more complex programs, use the PC application);
- data logging capabilites : storing information on usb flash drive, or sending it to PC terminal software;

3. Control measurement points for current - allow for measurement directly from the current measurement shunt;
4. Control measurement points for voltage - allows for measurement directly from battery voltage measurement cables;
5. Sockets for data logging and programming the display :
- **PC Conn :** for connection to PC, and programming;
- **USB Drive** : for logging process data on an usb flash drive, and possible test program uploading.

6. Screws and plates for fixing the tester to the cabinet.

MAIN SCREEN

Main screen is used for indicating the current state of the test process and its main parameters.
The screen looks as on the following image :



Indicators :

- **RUN : green** indicates test is working and output is enabled. **Gray** indicates test is stopped or finished, and output is disabled;
- **USB : green** indicates flash drive is connected and is ready to for reading and writing data. **Yellow** means drive is successfully recognized but is not ready, most often because it is formatted in NTFS. Note : drive must be FAT or FAT32 formatted.
- **PRG :** indicates different program states :
  - program is loaded and ready to be started : **yellow;**
  - program is working : **green;**
  - program is paused/stopped : **blue;**
  - program is not loaded or finished : **gray;**
- **CONN** : there is connection with the the PC application software via RS485 interface;
- **PAR :** the device is working in parallel.

Note : in parallel mode the device with the least address is a "master", all the rest are "slaves". For the slaves the "Program No. / Step No." text is replaced with "<SLAVE>".

Also, the "Current" indicates the total current for all the paralleled into group devices.

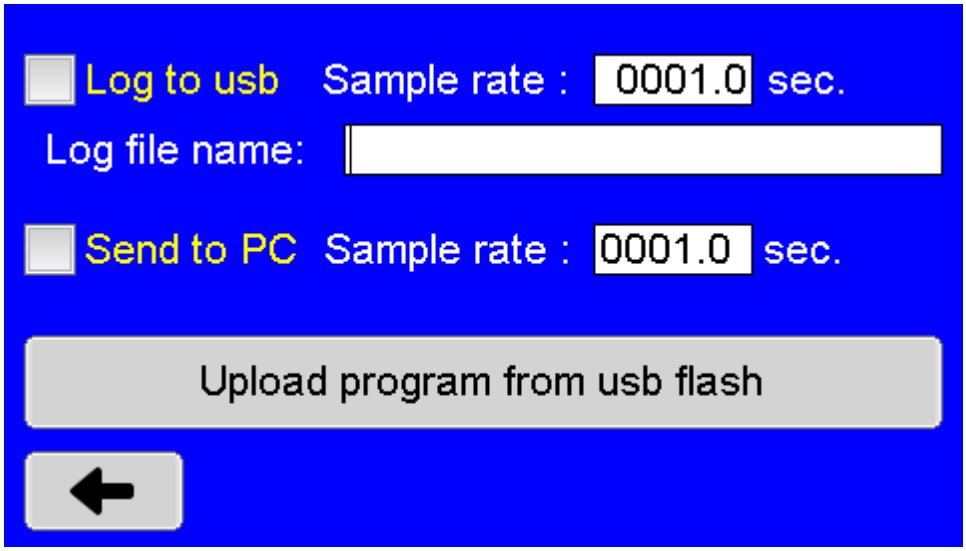
Buttons :

1. **START/STOP/Cont.** button : has an effect if there is loaded program. Starts, stops, resumes (from current or chosen step) a test program;
2. Opens a screen for activating data logging, or uploading a test program from flash drive;
3. "StLim" - opens the current program step's limits (current, voltage, power, sample time). On this screen, there is also the "END" button for ending the currently working test program;
4. Calls for the "SERVICE FUNCTIONS" screen, which contains the following options :
    - Diagnostic info : internal states, inputs and outputs, heatsink temperature and power supply voltage in charge mode;
    - TEST - allows creating simplified test program used for calibration;
    - DATE/TIME : for adjusting date and time;
    - PARAMETERS : for different adjustments and configuration. Password protected, only for authorized personnel;
    - ADC SETTINGS : password protected, only for authorized personnel;
    - INIT MEMORY: password protected, only for authorized personnel;
5. Opens screen with additional test information :

- **Id** - device current. In single mode it coincides with the current shown on the main screen. In parallel mode shows the current only for the device.
- **Ure** - reference electrode voltage;
- **PT and ST :** program and step time;
- **AH and WH** : amper hours and watt hours;
- **CL** : cycle counter
- **L1, L2, L3** : loop counters;

LOG SETTINGS SCREEN

Log settings screen is used for activation of the two available data logging options. Also, it is a means for accessing the option for choosing a test program saved on an usb drive, and uploading it for subsequent execution.



Log to usb:

Activates the logging on the usb flash drive. The user can choose the rate at which the log data is saved, and an explicit file name. If file name is not entered, a default one is used. Also, the files are saved in a predefined folder called "TESTS_CDT50A_18V". The start time of the test is added to the file name.

Note : the log starts when the program starts and continues at the defined sample rate until it is finished.

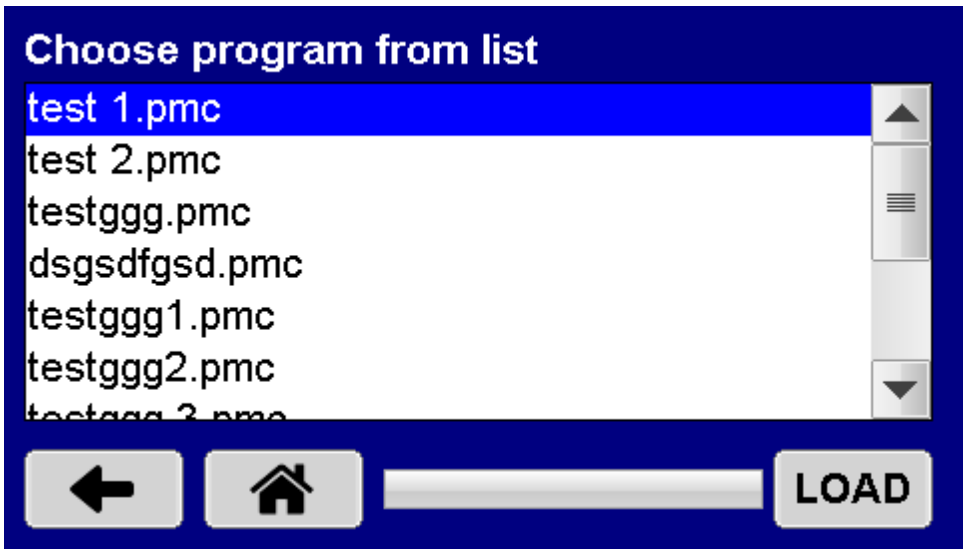
"USB drive" socket should be used for logging to USB.

Send to PC:

Activates sending the sample data to a PC via an USB 2.0 A cable. **A special driver from STMicroelectronics for virtual serial port creation should be installed.** A suitable terminal software should be used with the following serial port parameters :

- Baud rate : 921 600;
- Data bits : 8
- No parity
- Stop bits : 1
- Flow control : None.

"UPLOAD PROGRAM FROM USB FLASH" : opens the following screen for choosing a program from a list :



Note : the test programs are exported from the "Battery Test Manager" PC software. There is a strict rule where they are saved : in a folder named "CDT50". After a program is selected and the "LOAD" button is clicked, the program is loaded and the main screen is shown for the user to start the program.

For returning to the previous screen, the user clicks : 

ALARM MESAGES he main screen, the button is :  .

While test is working, if a failure happens, the respective protection is activated, and the output is disabled. The test program is automatically stopped and an alarm message with red background is shown on the screen.

The alarm messages are the following :

"Deviation from reference current!"
"Deviation from reference voltage!"
"Deviation from reference power!"
"OVER CURRENT!"
"OVER VOLTAGE!"
"OVERHEAT"
"POWER CABLES POLARITY REVERSED"

The alarm messages are cleared when program is resumed, for the exception of the reversed polarity of the power cables.

If the tests stops with "OVERHEAT", wait for the tester to cool off, before trying to start it again.

OPERATION

Before users start to work with the testers, it is highly recommended that they are familiar with this manual and with the relevant user manuals for the computer software.

The regular sequence of actions when using the devices is as follows :

1. First, connect the power output and measurement cables to the batteries. The tester must be switched off;
2. Switch on the power supply of the tester;
3. Testers work with test programs. The programs are created in the respective PC application software. The users should create test programs, and subsequently load a chosen program from the PC to the device. Also, the user can export the programs to an usb drive, which she can execute directly from the tester without the need of PC connection. In this case the data log options should be used. See topic "LOG SETTINGS SCREEN" for more information.
4. After a program is loaded the "PRG" indicator is colored **yellow**. This means that the program is ready to be started, so the next step is to actually start it. **If data log options are not activated the first start of a test program must be from the PC software!**
   After the program is started the "PRG" indicator becomes **green**.
5. The execution of a program can be suspended by an operator or when there is a protection fault. The "RUN" indicator is colored in **grey**. The "PRG" indicator becomes **blue**.
6. The program can be resumed after the reasons for its stopping are clarified and the problems solved. The resume can be done either from the PC or from the tester with the "Cont." button. **A program can be resumed from the current (from the interrupt point) or another step**. Again, both "RUN" and "PRG" indicators are **green**.
   Note : the testers support auto restart of the test program after restoring of broken power supply.
7. A program can be ended prematurely if needed. This can be done either from the PC or from the tester (from the step limits screen).
8. If a change of batteries is needed, the operator should first switch off the tester. Then he/she can start new test following the aforementioned steps.

Note : if during the test a correction in program is needed, the user can pause the test, make the corrections in the program and upload it again. Then test can be resumed either from the interrupt point or at different step.

MAINTENANCE 8

INTRODUCTION

This Chapter provides information about maintenance, calibration and troubleshooting.

UNITS UNDER WARRANTY

Units requiring repair during the warranty period should be returned to Pi Em Si Ltd. service facility. Unauthorized repairs performed by other than the company service facilities will void the warranty.

PERIODIC MAINTENANCE

No routine maintenance of the testers is required except for periodic cleaning. To clean, disconnect the unit from the AC supply and allow 60 seconds for discharging internal voltage. The metal surfaces should be cleaned using a damp cloth containing a mild solution of detergent and water. The solution should be applied onto a soft cloth, and not directly to the surface of the unit. Do not use aromatic hydrocarbons or chlorinated solvents for cleaning. Use low pressure compressed air to blow dust from the unit surface.

ADJUSTMENTS AND CALIBRATION

No internal adjustment or calibration is required. There is NO REASON to open the unit cover. **This will void the warranty!**

It is recommended to annually check the equipment and if adjustments and calibrations are needed they may be performed only by specialists of Pi Em Si Ltd.

For quick check, the control measurement point at the front face of testers can be used. For more detailed check, external calibrated shunt can be used.

PARTS REPLACEMENTS AND REPAIRS

As repairs are made only by the manufacturer, no parts replacement information is provided in the manual. In case of failure, unusual or erratic operation of the unit, contact the manufacturer.

SOFTWARE UPGRADE

Software upgrades are free in the warranty period, and can be done by the user under manufacturer guidance. The manufacturer gives the necessary files, drivers and instructions.

TROUBLESHOOTING

If a charge discharge tester appears to be operating improperly, use this troubleshooting guide to determine whether the power supply, load or external control circuit are the cause.
The table below provides the basic checks that can be performed to diagnose problems, and suggests certain actions that should be taken.

SYMPTOM	CHECK	ACTION
The TFT touch display is blank.	A. Is the AC power cord unplugged? B. If there is power supply to the tester.	A. Plug the power cord. B. Switch the fuses on. If the fuses turn off again, contact the testers manufacturer. C. Contact the manufacturer for repair.
No connection to PC. • The shape describing the tester in the PC software is filled in white. • The "CONN"	A. Is the tester power supplied? If yes, is it switched on ? B. Is the PC software application started?	A. Connect to power supply, as described in chapter "Installation". Switch on the unit. B. Start the application.

indicator on the main screen of the tester's display is grayed out.	C. Check if there are no LEDs lighted and/or blinking on the RS 485/232 converter. D. Check the cables between the RS 485/232 converter and the PC, and between the converter and the tester. Are all plugged in the right sockets?	C. Power supply the converter. D. Replace the converter's power adapter. E. Plug the cables in the right sockets.
The test state indicators of the tester's display are lit, but other indicators are not changing (time, current, voltage, etc.)	A. Check if in the PC software the program is working normally.	If everything is OK on the PC, wait for the test end and switch off and switch on the tester from the main power push button. Contact the manufacturer.
There are gaps or the logging on the PC software appears to has stopped.	A. Check the sample time in the test program steps. Too low sample time can cause samples loss.	A. Change the sample time.
There is no output current.	A. Are the power cables properly connected to the testers and to the batteries? B. Check your test program for correctness.	A. Connect the cables properly. Refer to "Preparation for use" topic in chapter "Installation" for more information. B.Correct the erroneous step parameters or limit conditions. Refer

		to the relevant software manual for more detail.
No measured voltage	A. Check how the measurement cables are connected. B. Check the measurement cables fuses.	A. Connect the measurement cables properly. B. If damaged, replace the fuses. The new ones must be with proper nominal. Damaged fuses are indication for damaged measurement cables, so they may have to be replaced too.
Tester gives alarm "OVERHEAT" in discharge mode and turns off.	A. Check the temperature in the room. B. Check if the fans of the testers are working.	A. Make sure there is normal ambient temperature. B.Contact the manufacturer.

If the users fail to diagnose the problem or the solutions described in this troubleshooting guide are not working, please contact the manufacturer.